Cybersecurity Lab Report: Implementing and Testing Wazuh SIEM

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Project Purpose: Learning SIEM deployment and validating real-time threat detection using MITRE ATT&CK.

Tools Used: VMware Workstation, Wazuh Manager (Ubuntu), Windows 11 (Victim), Kali Linux (Attacker).

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1. Project Overview

I built this lab to get hands-on experience with Security Information and Event Management (SIEM). My goal was to see how a real-world monitoring tool like **Wazuh** can detect "hacker" activities coming from a **Kali Linux** machine.

Lab Setup:

- **Infrastructure:** A virtual network with a Wazuh Server, a Windows target, and a Kali Linux attacker.

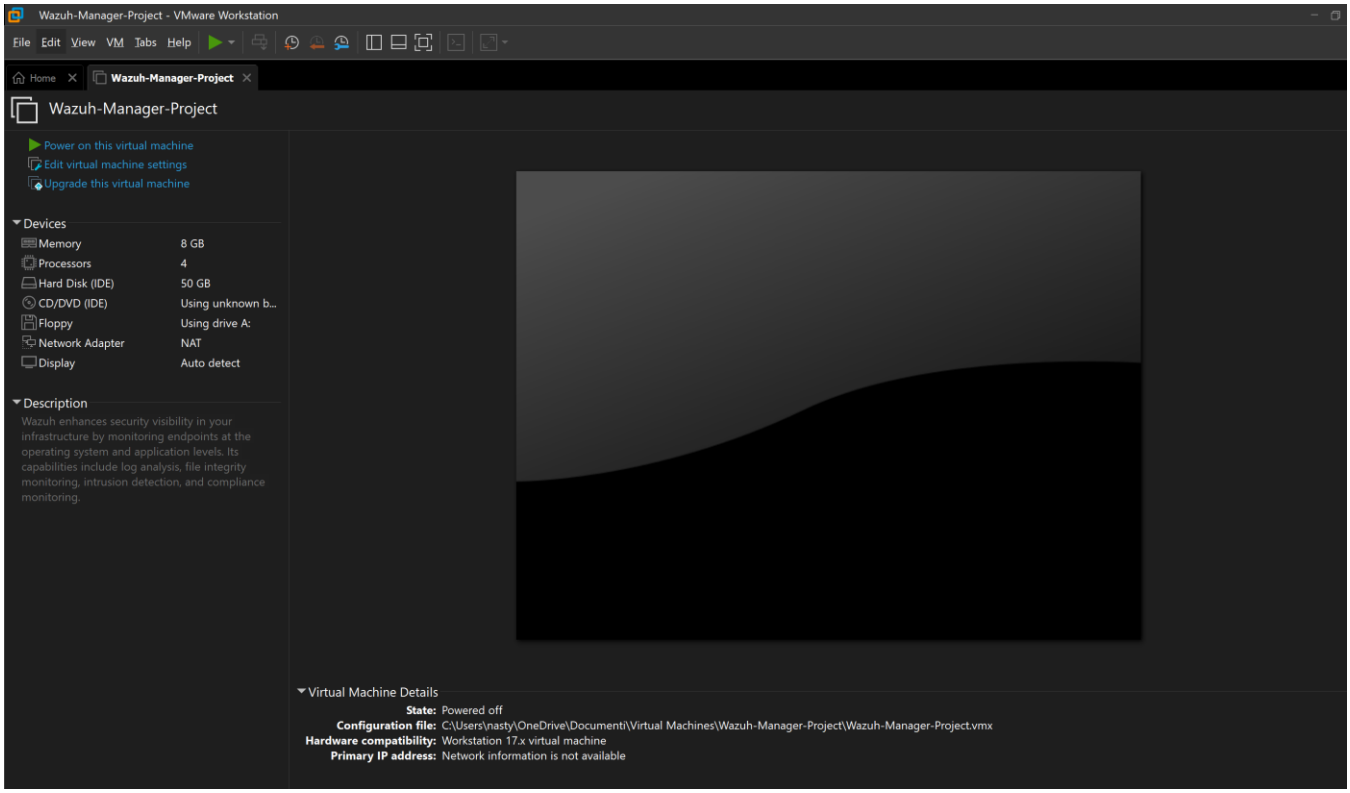

Figure 1: Virtual machine infrastructure setup in VMware Workstation.

- **Wazuh Manager:** The brain of the operation, running on Ubuntu at 192.168.208.129.
- **Wazuh Agent:** I installed this on my Windows 11 machine using a PowerShell script.

Figure 2: Automated Wazuh Agent installation on Windows 11 via PowerShell.

- It showing as **"Active"** in the dashboard.

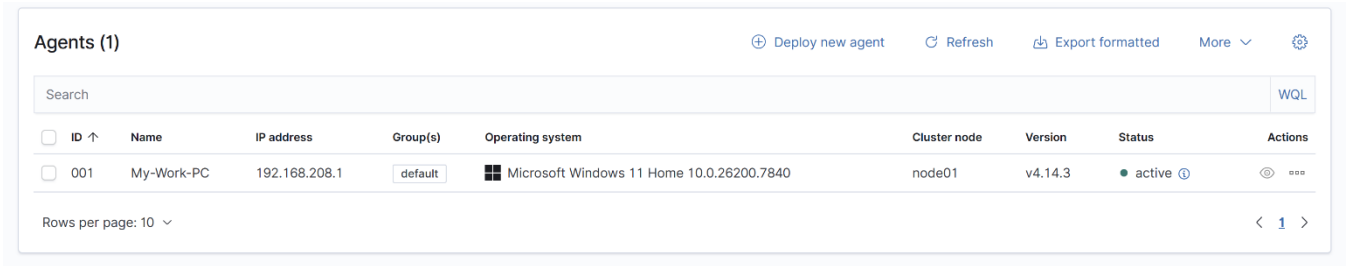

Figure 3: Verification of the Wazuh Agent's active status in the management console.

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2. Attack Simulations & Detection Analysis

Phase 1: Network Discovery (Nmap)

I started by scanning my Windows machine from Kali Linux using **Nmap** to find open ports.

- **What I did:** Ran an aggressive scan: sudo nmap -A -v 192.168.208.1.
- **What Nmap found:** It correctly identified an open MySQL port (3306/tcp).

Figure 4: Nmap scan results identifying an open MySQL port (3306/tcp) on the target host.

- **Detection in Wazuh:** The SIEM noticed the scan. Even though the dashboard auto-mapped some events to Defense Evasion, I analyzed the logs and realized this activity actually belongs to **Discovery (T1046)** in the MITRE framework.

Phase 2: Brute-Force Simulation (Hydra)

Once I found the MySQL port, I tried a "dictionary attack" using **Hydra**.

- **What happened:** I ran hydra -l root -p 123456 192.168.208.1 mysql, but the attack was blocked with **Error 1130**.

```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -p 123456 192.168.208.1 mysql
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-
binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-01 11:04:23
[INFO] Reduced number of tasks to 4 (mysql does not like many parallel connections)
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking mysql://192.168.208.1:3306/
[ERROR] Host '192.168.208.128' is not allowed to connect to this MySQL server
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-01 11:04:24
```

Figure 5: Failed brute-force attempt using Hydra, mitigated by secure MySQL host-access restrictions.

- **Lesson learned:** The MySQL server was configured to deny remote connections from my Kali IP. This was a great example of a "fail-safe" configuration. Wazuh captured this as a network anomaly rather than a specific login failure, because the connection was killed before it could even try the password.

Phase 3: Critical Account Activity

I manually failed several login attempts on my Windows machine to see if Wazuh would catch me.

Results: The dashboard lit up immediately! I saw **Level 5** alerts for the failed logins and **Level 8** alerts for changing account settings.

MITRE Mapping (My Corrections): While looking at the results, I noticed that the SIEM's automatic mapping wasn't 100% accurate, so I had to fix it manually:

- **From "Impact" to "Credential Access":** The dashboard tagged my failed logins as **Impact (T1531)**. But I learned that T1531 is actually about deleting or messing up accounts to cause damage. Since I was just trying to get in with the wrong password, I corrected this to **Credential Access (T1110 – Brute Force)**.
- **Refining "Persistence":** The system flagged my account changes under the general category of **Persistence**. I dug a bit deeper and mapped it specifically to **Account Manipulation (T1098)**. This makes more sense because it shows exactly how a hacker tries to change account permissions to stay in the system for a long time.

Mar 1, 2026 @ 17:32:00.018	My-Work-PC	T1098	Persistence	User account changed	8	6011
Mar 1, 2026 @ 17:31:38.746	My-Work-PC	T1531	Impact	Logon Failure - Unknown user or bad password	5	6012
Mar 1, 2026 @ 17:31:36.853	My-Work-PC	T1531	Impact	Logon Failure - Unknown user or bad password	5	6012
Mar 1, 2026 @ 17:31:33.474	My-Work-PC	T1531	Impact	Logon Failure - Unknown user or bad password	5	6012

Figure 6: Real-time security alerts categorized by severity (Level 5/8) and corrected MITRE ATT&CK tactics.

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3. The "Big Lesson": Troubleshooting Time Drift

This was the most challenging but interesting part of the project. At first, my attacks weren't showing up in the "Last 15 minutes" view.

- **Discovery:** I compared my Kali terminal time (11:09) with my Windows clock (17:11) and found a **6-hour difference**.
- **The Problem:** SIEM systems rely on perfect time synchronization to correlate events. My logs from the Windows Agent were being ingested, but they didn't match the "attacker's timeline" because of the clock skew.
- **The Fix:** I reconfigured the time zone on Kali Linux and enabled **NTP synchronization**.
- **Why it matters:** In a real job, if your clocks aren't synced, you can't prove *when* or *how* an attack happened.

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4. Final Thoughts

Building this lab taught me that SIEM is more than just looking at graphs. It’s about understanding the logic behind the logs and knowing how to fix technical issues like time drift to keep the data accurate. I’m now much more comfortable using **Wazuh** and **Kali** together for security monitoring.